

# The **exonot** package\*

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## Abstract

**exonot** provides standardized macros for exoplanet science and planetary astrophysics notation: masses and radii, system designations, atmospheric and spectroscopy symbols, orbital elements, and observational and statistical quantities. Every macro is wrapped in `\ensuremath`, so it works in both text and math mode, and all subscripts are typeset upright (`\mathrm`) following IAU style.

## 1 Usage

Load the package in the preamble:

```
\usepackage{exonot}    default
\usepackage[symbols]{exonot}    also defines \Earth and \Sun
\usepackage[siunitx]{exonot}    defines SI units if siunitx is loaded
first
```

`\Mjup` Quantity macros expand to a symbol with an upright subscript, e.g. `\Mjup`  $\rightarrow M_J$ , `\Rearth`  $\rightarrow R_\oplus$ , `\Teq`  $\rightarrow T_{\text{eq}}$ .

`\Teq` System notation: `\star{A}`  $\rightarrow A_\star$  is a host-star/component label carrying a star subscript, `\planet{b}`  $\rightarrow b$  is an upright planet letter, and `\planet \system{WASP-39}{b}`  $\rightarrow$  WASP-39 b typesets a full system name with correct `\system` non-breaking spacing.

`\vmr` A few macros take arguments: `\vmr{\H0}`  $\rightarrow X_{\text{H}_2\text{O}}$  is a volume mixing ratio, `\ldc` and `\ldc` takes an optional index, so `\ldc`  $\rightarrow u$  while `\ldc[1]`  $\rightarrow u_1$ .

## 2 Package options

**symbols** Loads `textcomp` and defines `\Earth` ( $\oplus$ ) and `\Sun` ( $\odot$ ) for use in text or math.

**siunitx** If `siunitx` is already loaded, declares the units `\earthmass`, `\jupitermass`, `\solarmass`, `\earthradius`, `\jupiterradius`, `\solarradius` and `\solarluminosity`; otherwise it issues a warning rather than failing.

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\*This document corresponds to **exonot** v1.1.0, dated 2026/06/27.

### 3 A note on `\star`

In standard L<sup>A</sup>T<sub>E</sub>X `\star` is the  $\star$  binary operator. `exonot` redefines it for the `\starsym` system-notation API `\star{name}`. The original symbol is preserved as `\starsym` in case the operator is still needed.

## 4 Implementation

### 4.1 Identification

```
1 \<package>
2 \NeedsTeXFormat{LaTeX2e}[2020/10/01]
3 \ProvidesPackage{exonot}[2026/06/27 v1.1.0 Standardized exoplanet and planetary astrophysics]
```

### 4.2 Package options

Two boolean flags, populated by the option declarations.

```
4 \newif\ifexo@symbols
5 \newif\ifexo@siunitx
6 \DeclareOption{symbols}{\exo@symbolstrue}
7 \DeclareOption{siunitx}{\exo@siunitxtrue}
8 \DeclareOption*{\PackageWarning{exonot}{Unknown option '\CurrentOption'}}
9 \ProcessOptions\relax
```

### 4.3 Required packages and `\star` preservation

We need `amsmath` and `amssymb`, and we stash the original `\star` operator before it is redefined.

```
10 \RequirePackage{amsmath}
11 \RequirePackage{amssymb}
12 \let\exo@starsym\star
13 \providecommand{\starsym}{\exo@starsym}
```

### 4.4 Units and physical quantities

Masses, radii, luminosity, temperatures, gravity, period, axis, transit depth and scale height.

```
14 \newcommand{\Mearth}{\ensuremath{M_{\oplus}}}} % M with Earth symbol -> M_Earth
15 \newcommand{\Mjup}{\ensuremath{M_{\mathrm{J}}}}} % M with roman "J" -> M_J
16 \newcommand{\Msun}{\ensuremath{M_{\odot}}}} % M with Sun symbol -> M_Sun
17 \newcommand{\Rearth}{\ensuremath{R_{\oplus}}}} % R_Earth
18 \newcommand{\Rjup}{\ensuremath{R_{\mathrm{J}}}}} % R_J
19 \newcommand{\Rsun}{\ensuremath{R_{\odot}}}} % R_Sun
20 \newcommand{\Lsun}{\ensuremath{L_{\odot}}}} % L_Sun
21 \newcommand{\Teff}{\ensuremath{T_{\mathrm{eff}}}}} % T_eff
22 \newcommand{\logg}{\ensuremath{\log g}} % log g
23 \newcommand{\Porb}{\ensuremath{P_{\mathrm{orb}}}}} % P_orb
24 \newcommand{\aorb}{\ensuremath{a}} % a
25 \newcommand{\Teq}{\ensuremath{T_{\mathrm{eq}}}}} % T_eq
26 \newcommand{\tdepth}{\ensuremath{\left(R_{\mathrm{p}}/R_{\exo@starsym}\right)^2}} % (R_p/R_*)^2
27 \newcommand{\Hatm}{\ensuremath{H}} % H
```

Generic planet and star quantities (the stellar versions reuse the preserved star symbol as subscript).

```

28 \newcommand{\Mp}{\ensuremath{M_{\mathrm{p}}}} % planet mass -> M_p
29 \newcommand{\Rp}{\ensuremath{R_{\mathrm{p}}}} % planet radius -> R_p
30 \newcommand{\Mstar}{\ensuremath{M_{\mathrm{exo@starsym}}}} % stellar mass -> M_star
31 \newcommand{\Rstar}{\ensuremath{R_{\mathrm{exo@starsym}}}} % stellar radius -> R_star
32 \newcommand{\rho_p}{\ensuremath{\rho_{\mathrm{p}}}} % planet density -> rho_p
33 \newcommand{\rho_star}{\ensuremath{\rho_{\mathrm{exo@starsym}}}} % stellar dens.-> rho_star

```

Physical constants.

```

34 \newcommand{\kB}{\ensuremath{k_{\mathrm{B}}}} % Boltzmann constant -> k_B
35 \newcommand{\mH}{\ensuremath{m_{\mathrm{H}}}} % hydrogen-atom mass -> m_H
36 \newcommand{\amu}{\ensuremath{m_{\mathrm{u}}}} % atomic mass unit -> m_u
37 \newcommand{\Gnewton}{\ensuremath{G}} % gravitational constant -> G
38 \newcommand{\stefan}{\ensuremath{\sigma_{\mathrm{SB}}}} % Stefan-Boltzmann -> sigma_SB

```

## 4.5 Exoplanet system notation

```

39 \renewcommand{\star}[1]{\ensuremath{\#1_{\mathrm{exo@starsym}}}} % \star{A} -> A_star
40 \newcommand{\planet}[1]{\textnormal{\#1}} % \planet{b} -> b
41 \newcommand{\system}[2]{\#1\nobreakspace\planet{\#2}} % \system{WASP-39}{b} -> WASP-39 b

```

## 4.6 Atmospheric and spectroscopy notation

Common molecules. Macro names follow the chemical formula: monoxides are `\CO`, `\SO`, `\SiO`; dioxides are `\COtwo`, `\SOtwo`, `\SiOtwo`.

```

42 \newcommand{\H0}{\ensuremath{\mathrm{H}_2}} % water -> H2O
43 \newcommand{\CO}{\ensuremath{\mathrm{CO}}} % carbon monoxide -> CO
44 \newcommand{\COtwo}{\ensuremath{\mathrm{CO}_2}} % carbon dioxide -> CO2
45 \newcommand{\CH}{\ensuremath{\mathrm{CH}_4}} % methane -> CH4
46 \newcommand{\NH}{\ensuremath{\mathrm{NH}_3}} % ammonia -> NH3
47 \newcommand{\SO}{\ensuremath{\mathrm{SO}}} % sulfur monoxide -> SO
48 \newcommand{\SOtwo}{\ensuremath{\mathrm{SO}_2}} % sulfur dioxide -> SO2
49 \newcommand{\SiO}{\ensuremath{\mathrm{SiO}}} % silicon monoxide -> SiO
50 \newcommand{\SiOtwo}{\ensuremath{\mathrm{SiO}_2}} % silicon dioxide -> SiO2
51 \newcommand{\Otwo}{\ensuremath{\mathrm{O}_2}} % molecular oxygen -> O2
52 \newcommand{\Othree}{\ensuremath{\mathrm{O}_3}} % ozone -> O3
53 \newcommand{\Ntwo}{\ensuremath{\mathrm{N}_2}} % molecular nitrogen -> N2
54 \newcommand{\Htwo}{\ensuremath{\mathrm{H}_2}} % molecular hydrogen -> H2
55 \newcommand{\Helium}{\ensuremath{\mathrm{He}}} % helium -> He
56 \newcommand{\HtwoS}{\ensuremath{\mathrm{H}_2\mathrm{S}}} % hydrogen sulfide -> H2S
57 \newcommand{\HCN}{\ensuremath{\mathrm{HCN}}} % hydrogen cyanide -> HCN
58 \newcommand{\CtwoHtwo}{\ensuremath{\mathrm{C}_2\mathrm{H}_2}} % acetylene -> C2H2
59 \newcommand{\OCS}{\ensuremath{\mathrm{OCS}}} % carbonyl sulfide -> OCS
60 \newcommand{\PHthree}{\ensuremath{\mathrm{PH}_3}} % phosphine -> PH3

```

Metal oxides and hydrides (hot-Jupiter optical absorbers). Note that `\FeHyd` is the iron-hydride *molecule*; the metallicity  $[\mathrm{Fe}/\mathrm{H}]$  is `\FeH` below.

```

61 \newcommand{\TiO}{\ensuremath{\mathrm{TiO}}} % titanium oxide -> TiO
62 \newcommand{\VO}{\ensuremath{\mathrm{VO}}} % vanadium oxide -> VO
63 \newcommand{\FeHyd}{\ensuremath{\mathrm{FeH}}} % iron hydride -> FeH (cf. \FeH = [Fe/H])
64 \newcommand{\CaH}{\ensuremath{\mathrm{CaH}}} % calcium hydride -> CaH
65 \newcommand{\MgH}{\ensuremath{\mathrm{MgH}}} % magnesium hyd. -> MgH

```

Atomic and ionic absorbers; the ionisation stage is a small-caps roman numeral.

```

66 \newcommand{\NaI}{\ensuremath{\mathrm{Na}\,,\,\textsc{i}}} % neutral sodium -> Na I
67 \newcommand{\KI}{\ensuremath{\mathrm{K}\,,\,\textsc{i}}} % neutral potassium -> K I
68 \newcommand{\FeI}{\ensuremath{\mathrm{Fe}\,,\,\textsc{i}}} % neutral iron -> Fe I
69 \newcommand{\FeII}{\ensuremath{\mathrm{Fe}\,,\,\textsc{ii}}} % singly-ionised Fe -> Fe II
70 \newcommand{\Hminus}{\ensuremath{\mathrm{H}^{-}}} % H minus ion -> H^-
71 \newcommand{\eminus}{\ensuremath{\mathrm{e}^{-}}} % electron -> e^-

```

Mean molecular weight, volume mixing ratio, composition ratios and metallicity, spectral resolution, and stellar contamination.

```

72 \newcommand{\mmw}{\ensuremath{\mu}} % mu
73 \newcommand{\vmr}[1]{\ensuremath{X_{\#1}}} % volume mixing ratio
74 \newcommand{\CtoO}{\ensuremath{\mathrm{C/O}}} % carbon-to-oxygen ratio -> C/O
75 \newcommand{\FeH}{\ensuremath{\mathrm{Fe/H}}} % iron metallicity -> [Fe/H]
76 \newcommand{\MH}{\ensuremath{\mathrm{M/H}}} % bulk metallicity -> [M/H]
77 \newcommand{\Rspec}{\ensuremath{R_{\lambda}}} % R_lambda
78 \newcommand{\fcont}{\ensuremath{f_{\mathrm{cont}}}} % f_cont

```

Thermal structure.

```

79 \newcommand{\Tday}{\ensuremath{T_{\mathrm{day}}}} % dayside temperature -> T_day
80 \newcommand{\Tnight}{\ensuremath{T_{\mathrm{night}}}} % nightside temperature -> T_night
81 \newcommand{\Tint}{\ensuremath{T_{\mathrm{int}}}} % internal temperature -> T_int
82 \newcommand{\Tirr}{\ensuremath{T_{\mathrm{irr}}}} % irradiation temp. -> T_irr
83 \newcommand{\Tbright}{\ensuremath{T_{\mathrm{b}}}} % brightness temp. -> T_b
84 \newcommand{\opacity}{\ensuremath{\kappa}} % opacity -> kappa
85 \newcommand{\optdepth}{\ensuremath{\tau}} % optical depth -> tau
86 \newcommand{\albedo}{\ensuremath{A_{\mathrm{B}}}} % Bond albedo -> A_B
87 \newcommand{\insol}{\ensuremath{S}} % instellation flux -> S
88 \newcommand{\redist}{\ensuremath{f}} % heat-redistribution f -> f

```

## 4.7 Orbital and dynamical quantities

```

89 \newcommand{\ecc}{\ensuremath{e}} % eccentricity -> e
90 \newcommand{\inc}{\ensuremath{i}} % inclination -> i
91 \newcommand{\argperi}{\ensuremath{\omega}} % argument of periastron -> omega
92 \newcommand{\ascnode}{\ensuremath{\Omega}} % longitude of asc. node -> Omega
93 \newcommand{\Tc}{\ensuremath{T_{\mathrm{c}}}} % time of conjunction -> T_c
94 \newcommand{\Tzero}{\ensuremath{T_0}} % reference epoch -> T_0
95 \newcommand{\Tperi}{\ensuremath{T_{\mathrm{peri}}}} % time of periastron -> T_peri
96 \newcommand{\Krv}{\ensuremath{K}} % RV semi-amplitude -> K
97 \newcommand{\massfn}{\ensuremath{f(m)}} % spectroscopic mass fn. -> f(m)
98 \newcommand{\obliq}{\ensuremath{\lambda}} % sky-projected obliquity -> lambda

```

## 4.8 Observational quantities

```

99 \newcommand{\tdur}{\ensuremath{T_{14}}} % total transit duration -> T_14
100 \newcommand{\ting}{\ensuremath{T_{12}}} % ingress/egress duration -> T_12
101 \newcommand{\bparam}{\ensuremath{b}} % impact parameter -> b
102 \newcommand{\ldc}[1][\]{\ensuremath{u_{\#1}}} % limb-darkening coeff. -> u, u_1
103 \newcommand{\snr}{\ensuremath{S/N}} % S/N
104 \newcommand{\fratio}{\ensuremath{F_{\mathrm{p}}/F_{\mathrm{exo@starsym}}}} % planet/star flux -> F_p/F_star
105 \newcommand{\edepth}{\ensuremath{\delta_{\mathrm{ecl}}}} % eclipse depth -> delta_ecl
106 \newcommand{\ppm}{\ensuremath{\mathrm{ppm}}} % parts per million (unit) -> ppm

```

## 4.9 Statistics and model comparison

```

107 \newcommand{\chisq}{\ensuremath{\chi^2}} % chi-squared -> chi^2

```

```

108 \newcommand{\redchisq}{\ensuremath{\chi^2_{\nu}}} % reduced chi-squared -> chi^2_nu
109 \newcommand{\BIC}{\ensuremath{\mathrm{BIC}}} % Bayesian info crit. -> BIC
110 \newcommand{\AIC}{\ensuremath{\mathrm{AIC}}} % Akaike info crit. -> AIC
111 \newcommand{\lnL}{\ensuremath{\ln\mathrm{L}}} % log-likelihood -> ln L
112 \newcommand{\Bfac}{\ensuremath{\mathrm{B}}} % Bayes factor -> B

```

## 4.10 Optional symbols support

```

113 \ifexo@symbols
114   \RequirePackage{textcomp}
115   \providecommand{\Earth}{\ensuremath{\oplus}} % Earth symbol, text or math
116   \providecommand{\Sun}{\ensuremath{\odot}} % Sun symbol, text or math
117 \fi

```

## 4.11 Optional siunitx support

Units are declared only if siunitx is actually loaded, otherwise we warn instead of failing. The new names avoid clobbering the macros above.

```

118 \ifexo@siunitx
119   \AtBeginDocument{%
120     \@ifpackageloaded{siunitx}{%
121       \DeclareSIUnit\earthmass{\ensuremath{M_{\oplus}}}%
122       \DeclareSIUnit\jupitermass{\ensuremath{M_{\mathrm{J}}}}%
123       \DeclareSIUnit\solarmass{\ensuremath{M_{\odot}}}%
124       \DeclareSIUnit\earthradius{\ensuremath{R_{\oplus}}}%
125       \DeclareSIUnit\jupiterradius{\ensuremath{R_{\mathrm{J}}}}%
126       \DeclareSIUnit\solarradius{\ensuremath{R_{\odot}}}%
127       \DeclareSIUnit\solarluminosity{\ensuremath{L_{\odot}}}%
128     }{%
129       \PackageWarning{exonot}%
130         {Option 'siunitx' was requested but the siunitx package is not}%
131       \MessageBreak loaded; SI unit commands were not defined}%
132     }%
133   }%
134 \fi
135 \end{package}

```